

Homes and hiding

STAYING HIDDEN is one of the best means of defense—if a predator cannot see you, it cannot eat you! Many sea animals shelter among seaweeds, in rocky crevices, or under the sand. Matching the colors and even the texture of the background also helps sea creatures remain undetected. The sargassum fish even looks like bits of seaweed. Hard shells are useful armor, at least giving protection from weak-jawed predators. Sea snails and clams make their own shells that cover the body. Crab and lobsters have outer shells, like suits of armor, covering the body and each jointed limb. The hermit crab is unusual because only the front part of the body and the legs are covered by a hard shell. Its abdomen is soft, so a hermit crab uses the empty shell of a sea snail to protect itself.

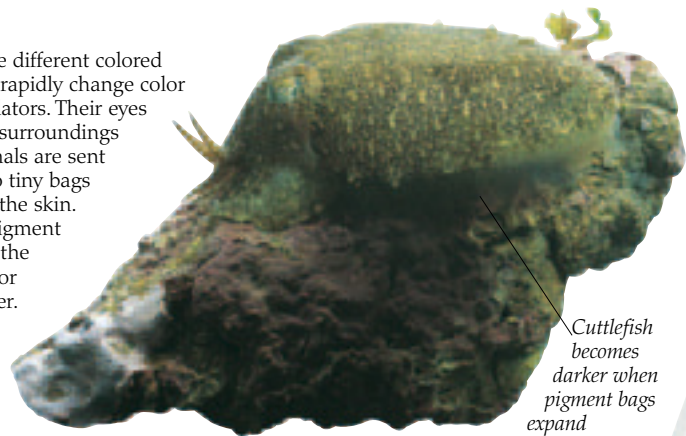


WHAT A WEED

This fish lives among floating clumps of sargassum seaweed, where frilly growths on its head, body, and fins help it avoid being seen by predators, making a realistic disguise. Many different animals live in sargassum seaweed, which drifts in large quantities in the Sargasso Sea of the North Atlantic.

BLENDING IN

Cuttlefish have different colored pigments and rapidly change color to escape predators. Their eyes perceive their surroundings and nerve signals are sent by the brain to tiny bags of pigment in the skin. When these pigment bags contract, the cuttlefish's color becomes lighter.



Cuttlefish becomes darker when pigment bags expand



Hermit crab leaving old whelk shell

Anemone

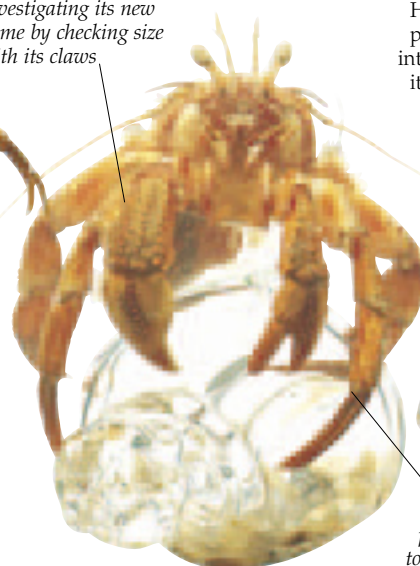
When out of its shell, crab is vulnerable to predators

Investigating its new home by checking size with its claws

Hermit crab can be persuaded to move into a perspex shell so its movements can be viewed

ALL CHANGE

Like all crustaceans, a hermit crab grows by shedding its hard, outer skeleton and does this within the safety of its snail shell home. As it grows larger, it needs to find a larger snail shell to move into. Before leaving its old shell, it will test the size of a new home. If it is not large enough or is cracked, the hermit crab looks for another shell. When the hermit crab has found one which is just right, it carefully pulls its body out of its old shell, tucking it quickly into the new shell. As the hermit crab grows larger it moves into large whelk shells and lives in shallow water submerged on the seabed.



Leg with pointed claws to get a grip on seabed when walking

Antenna